

**ΕΘΝΙΚΟ ΜΕΤΣΟΒΙΟ ΠΟΛΥΤΕΧΝΕΙΟ**
Σχολή Ηλεκτρολόγων Μηχανικών
και Μηχανικών Υπολογιστών

SaaS:
Τεχνολογίες Υπηρεσιών
Λογισμικού

# Cloud computing and the OpenStack use-case

Stamatis Katsaounis, Ph.D. student <skatsaounis@mail.ntua.gr>

# Agenda



# First Part – A flight above the Clouds

- Cloud Computing
- Cloud Providers
- OpenStack: The Open-Source Cloud
- OpenStack as an Operator
- OpenStack as a User

# Second Part – OpenStack usage

- Web Portal for Cloud resources
- OpenStack CLI
- Terraform resource provider

# A flight above the Clouds

First Part



# Cloud Computing

**What**

Globally distributed computing infrastructure

Computing services over the internet

Rental of CPU power and storage from someone else's data center

**Why**

High availability with geo redundancy

Disaster Recovery

Consumption-based model

Performance

# Cloud Models

**Public**

Cloud resources are owned and operated by a cloud service provider and delivered over the internet.

**Private**

Cloud resources can be physically located at on-premises datacenter or can be hosted by a service provider.

**Hybrid**

A combination of public and private cloud resources

# Cloud Services

**Infrastructure as a service (IaaS)**

Direct rental of IT infrastructure: servers and virtual machines (VMs), storage, networks, operating systems

**Platform as a service (PaaS)**

Cloud computing services that supply an on-demand environment for software applications (e.g., Managed Kubernetes)

**Software as a service (SaaS)**

Deliverable software applications over the Internet, on demand (e.g., Outlook, Gmail)


Google Cloud
IBM Cloud
Azure
DigitalOcean
aws
Cloud Providers
Alibaba Cloud

# Cloud in Research and Academia

- Open Clouds for Research Environments project (OCRE): https://www.ocre-project.eu/
- Cern Computing: https://home.cern/science/computing
- GEANT: https://www.geant.org/
- GRNET: https://grnet.gr/ , ~okeanos: https://okeanos.grnet.gr/







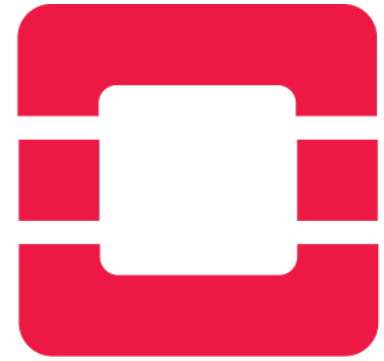





# OpenStack: The Open-Source Cloud

- Joined effort of Rackspace and NASA, created in 2010
- Entire code base is open-source
- Currently in 31st stable version, 2025.1 Epoxy
- Owned by Open Infrastructure Foundation and community

https://www.openstack.org/



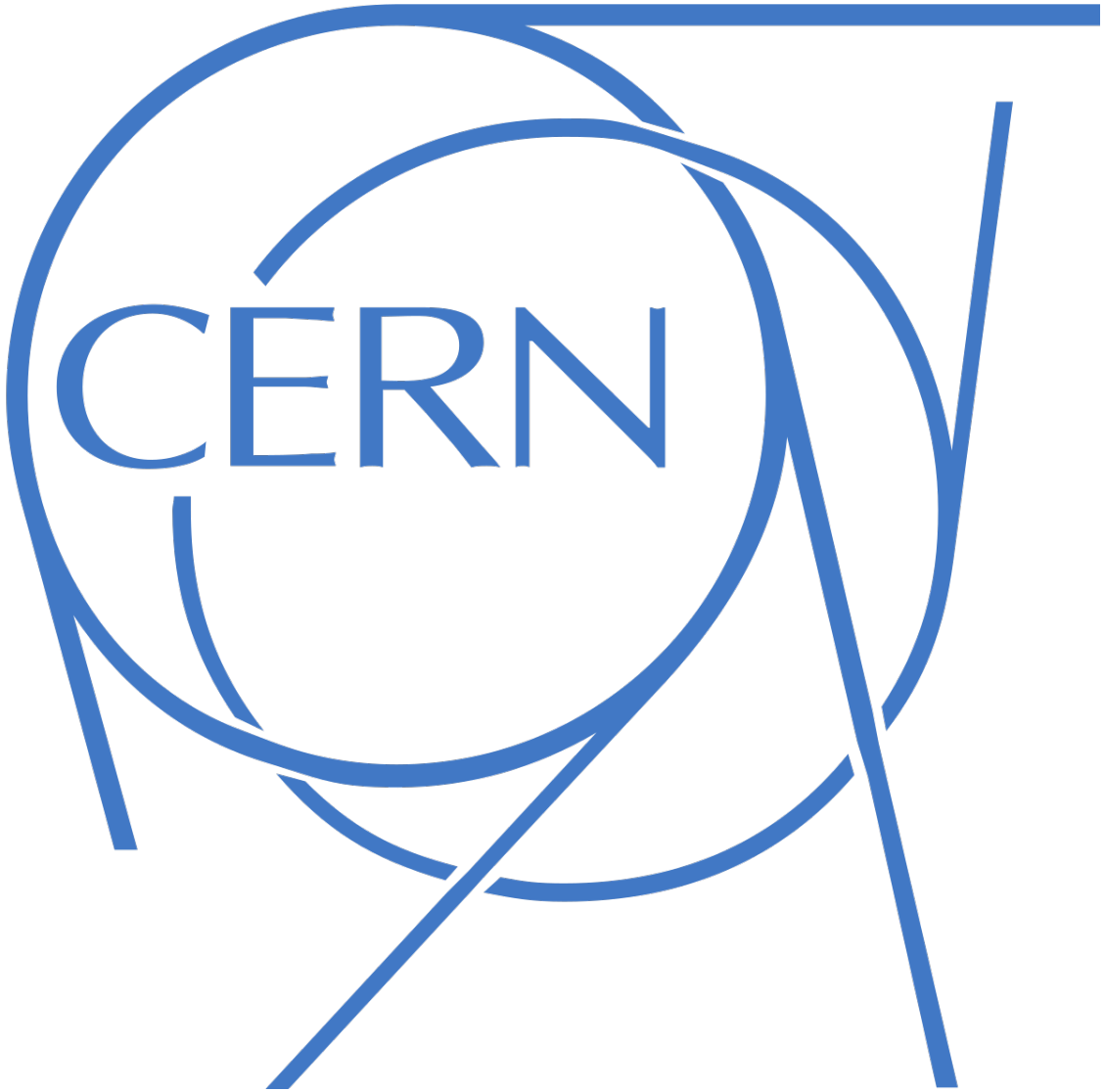
openstack®

# CERN Deployment

- 12k baremetal servers
- 360k CPU cores
- 2.22 PB of RAM

https://home.cern/science/computing

https://www.openstack.org/blog/10-years-of-openstack-tim-bell-at-cern/



# Other Use Cases

- Walmart
- Blizzard Entertainment
- Volkswagen Group
- Many more

https://www.openstack.org/use-cases/








# OpenStack @ Louros GRNET

- 150 baremetal servers
- 4.4k CPU cores
- 30 TB of RAM

https://www.youtube.com/watch?v=gUVAOWjo-OM



grnet

# OpenStack as an Operator

**Infrastructure Deployment**

Bare metal servers provisioning and configuration

Networking management and configuration

Storage provisioning

**Infrastructure Monitoring**

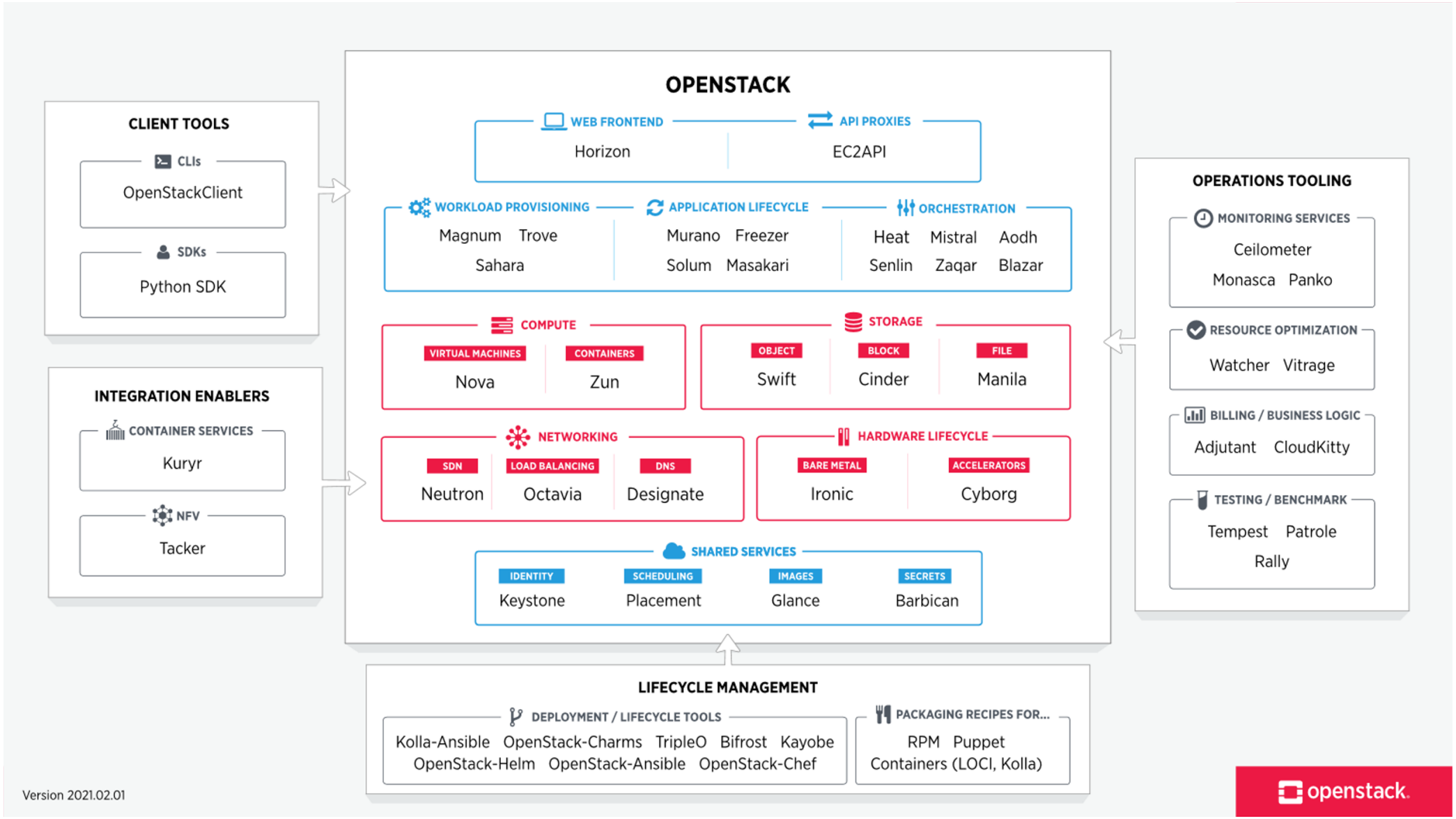
Server health and security

Networking Availability

**User and Quota Management**

User registration

Quotas assignment to projects


CLIENT TOOLS
CLIs
OpenStackClient
SDKs
Python SDK
INTEGRATION ENABLERS
CONTAINER SERVICES
Kuryr
NFV
Tacker
OPENSTACK
WEB FRONTEND
Horizon
API PROXIES
EC2API
WORKLOAD PROVISIONING
Magnum Trove
Sahara
APPLICATION LIFECYCLE
Murano Freezer
Solum Masakari
ORCHESTRATION
Heat Mistral Aodh
Senlin Zaqar Blazar
COMPUTE
VIRTUAL MACHINES
Nova
CONTAINERS
Zun
STORAGE
OBJECT
Swift
BLOCK
Cinder
FILE
Manila
NETWORKING
SDN
Neutron
LOAD BALANCING
Octavia
DNS
Designate
HARDWARE LIFECYCLE
BARE METAL
Ironic
ACCELERATORS
Cyborg
SHARED SERVICES
IDENTITY
Keystone
SCHEDULING
Placement
IMAGES
Glance
SECRETS
Barbican
OPERATIONS TOOLING
MONITORING SERVICES
Ceilometer
Monasca Panko
RESOURCE OPTIMIZATION
Watcher Vitrage
BILLING / BUSINESS LOGIC
Adjutant CloudKitty
TESTING / BENCHMARK
Tempest Patrole
Rally
LIFECYCLE MANAGEMENT
DEPLOYMENT / LIFECYCLE TOOLS
Kolla-Ansible OpenStack-Charms TripleO Bifrost Kayobe
OpenStack-Helm OpenStack-Ansible OpenStack-Chef
PACKAGING RECIPES FOR...
RPM Puppet
Containers (LOCI, Kolla)
Version 2021.02.01
openstack.

# OpenStack as a User

**Horizon - Web Portal for Cloud resources**

Virtual machine management

Private networking and public IP assignments

Storage management

**OpenStack CLI**

Lightweight command line for resource management

Can perform all portal actions and many more

Requires better expertise

**Infrastructure as Code – Terraform resource provider**

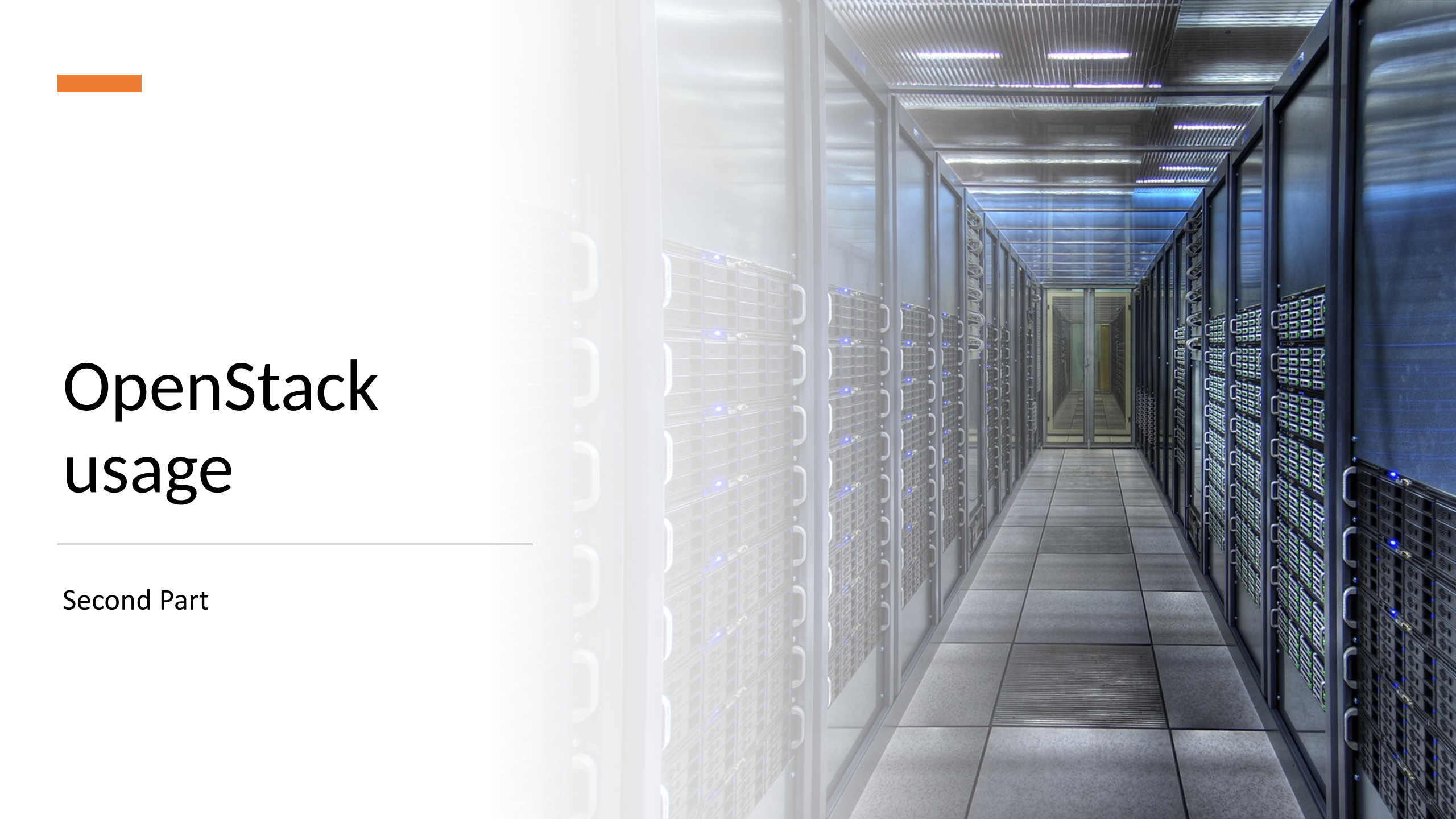
State of the art resource management

Used for production deployments

Declarative syntax

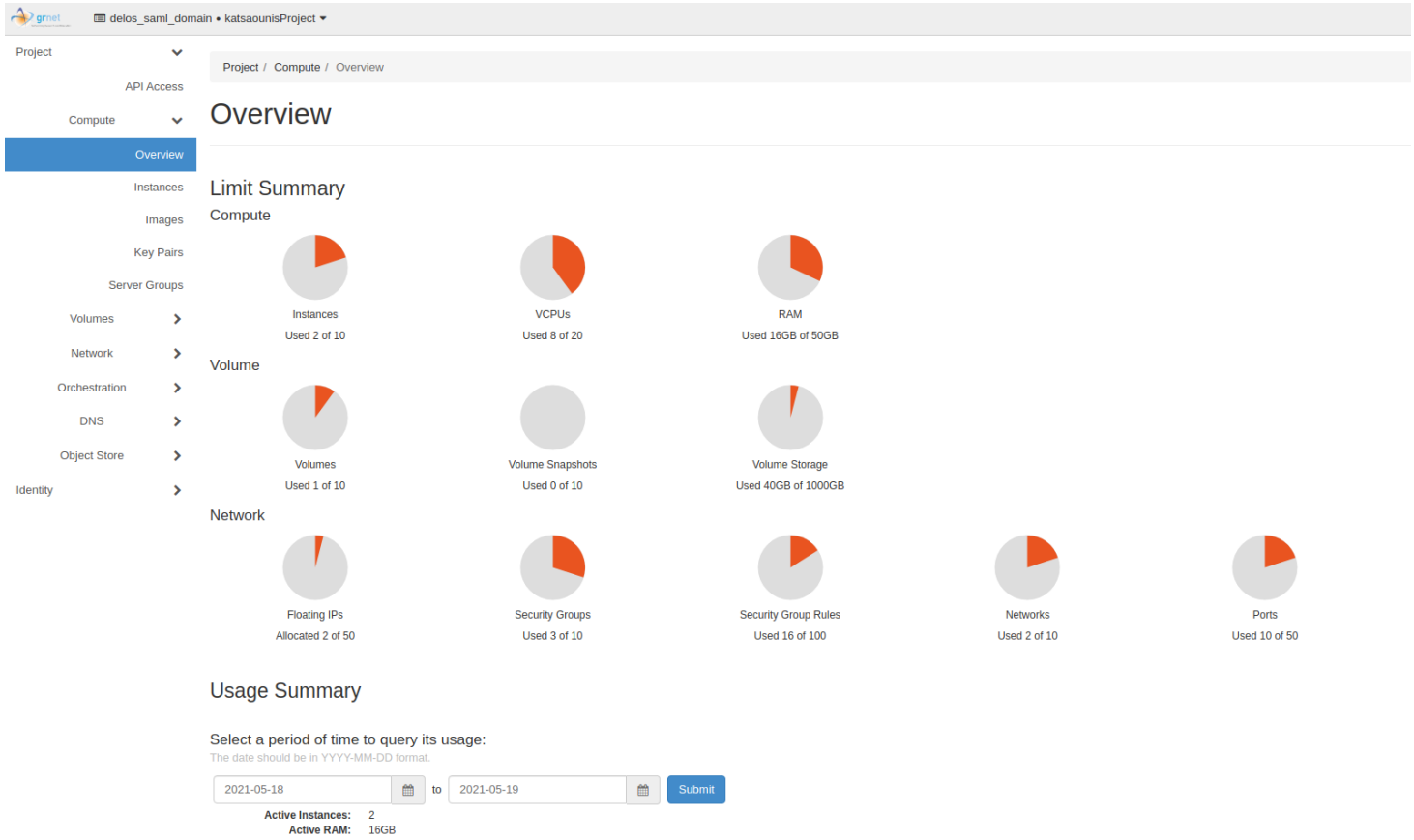
# OpenStack usage

Second Part



# Web Portal for Cloud resources

- Manage Virtual Machines
- Manage Networking
- Manage Storage
- Many other actions



# OpenStack CLI

- Manage Virtual Machines
- Manage Networking
- Manage Storage
- Many other actions

```
skatsaounis@XPS-13-9360:~/projects/ntua-phd/ntua-demo/terraform$ openstack server list
+--------------------------------------+----------+--------+-------------------------------------+--------------+--------+
| ID                                   | Name     | Status | Networks                            | Image        | Flavor |
+--------------------------------------+----------+--------+-------------------------------------+--------------+--------+
| 3fd85c5b-b0c2-4a3e-9c72-2c70d6b6dc53 | demo-k8s | ACTIVE | private-net=20.0.3.38, 62.217.83.220 | Ubuntu 20.04 | Gray   |
| 420d3dd9-19d2-4364-958e-7204ec17dc9e | test-vm  | ACTIVE | test-net=10.0.0.32, 62.217.83.60     |              | Gray   |
+--------------------------------------+----------+--------+-------------------------------------+--------------+--------+
skatsaounis@XPS-13-9360:~/projects/ntua-phd/ntua-demo/terraform$ openstack network list
+--------------------------------------+-------------------+--------------------------------------------------------------------------+
| ID                                   | Name              | Subnets                                                                  |
+--------------------------------------+-------------------+--------------------------------------------------------------------------+
| 00a0a75c-784a-4b37-a070-6525c4cbfba5 | private-net       | 19ad9f13-bb50-4c73-9ae0-dac122d0f247                                     |
| b60de9a1-9056-4952-a113-56c7497d9de8 | Direct-Internet   | 2784681b-2748-4c48-93ef-60219d824b79                                     |
| bd0999bf-7ed0-4c6e-81b3-f8f79acdae36 | test-net          | 53302b82-948d-447e-be06-d46f056cecc9                                     |
| fa87edbd-b40c-4144-b317-5838aaf440db | FloatingIP-Network | 33d9132e-cf13-4460-8074-a0d2b7c98d18, e6af1914-62f0-4b1e-aefd-0d23a4df6ea3 |
+--------------------------------------+-------------------+--------------------------------------------------------------------------+
```

# Terraform resource provider

- Describe resources in declarative way
- Provide credentials
- Declare implicit and explicit relationships between resources
- Terraform creates the resources by determining the correct order of operations

```
resource "openstack_networking_network_v2" "private_net" {
  name           = "private-net"
  admin_state_up = "true"
}

resource "openstack_networking_subnet_v2" "private_subnet" {
  name            = "private-subnet"
  network_id      = openstack_networking_network_v2.private_net.id
  cidr            = "20.0.0.0/16"
  ip_version      = 4
  dns_nameservers = ["8.8.8.8"]
}

resource "openstack_networking_router_v2" "net_router" {
  name                = "net-router"
  admin_state_up      = "true"
  external_network_id = data.openstack_networking_network_v2.os_extnet.id
}

resource "openstack_networking_router_interface_v2" "private_net_router_iface" {
  router_id = openstack_networking_router_v2.net_router.id
  subnet_id = openstack_networking_subnet_v2.private_subnet.id
}

```

# Q&A Section

"The important thing is not to stop questioning." - A. Einstein



# Thank you

